

Lucas Webster

Davis, CA | lucasfw@icloud.com | <https://lfw598.github.io/> | [linkedin.com/in/lucas-webster](https://www.linkedin.com/in/lucas-webster)
github.com/LFW598

Career Objectives

Statistics and Data Science is my interest, but helping people is my passion. I take pride in being able to work through mazes of data with statistics, especially when the outcome is the safety or treatment of real people.

Education

University of California, Davis, Ph.D. in Biostatistics Sept 2025 – Current

- **Overall GPA:** 3.92/4.0
- **Coursework:** Survival Analysis, Generalized Linear Models, Longitudinal Data, Computational Statistics

University of California, Santa Barbara, BS in Statistics and Data Science, Minor in Religious Studies Sept 2022 – June 2025

- **Overall GPA:** 4.0/4.0, Major GPA: 4.0/4.0
- **Coursework:** Experimental Design, Methods of Analysis, Linear Algebra, Stochastic Processes, SAS Base Programming, Bayesian Analysis

Experience

ISI-BUDS (SIBS), UC Irvine – Irvine, CA June 2024 – August 2024

- Learned and practiced the research pipeline for a biostatistics data analysis paper
- Presented at a public symposium on mhealth data and its relation to diabetes and hypertension in ethnic minority women
- Undertook a group research project over six weeks which culminated in the submission of a research paper

Statistics Undergraduate Learning Assistant, UC Santa Barbara April 2024 – June 2025

- Held office hours for students in intermediate statistics and data science courses
- Held review sessions to 100+ students before examinations
- Created over 100 problems of review material

Soccer Intramural Head of Officials, UC Santa Barbara October 2022 – June 2025

- Scheduled Referees for 30+ games every week
- Resolved disciplinary issues between players and staff
- On-boarded and trained 30+ referees

MAX Undergraduate Learning Assistant, UC Santa Barbara July 2024 – December 2024

- Guided incoming STEM freshman through universal STEM topics
- Served as a mentor to freshman on navigating the STEM college experience

Volunteer E3 Learning Tutor, Goleta Valley Junior High – Goleta, CA September 2023 – March 2024

- Volunteered weekly at the local junior high school tutoring math
- Worked with five students who were assessed to need individual help
- Saw test scores improve in all students worked with

Data Writer, The Daily Nexus – Goleta, CA October 2022 – October 2023

- Wrote articles published to the print and digital versions of the Daily Nexus Newspaper
- Interviewed subjects of articles
- Collected data and made visualizations for articles

Projects

Liver Steatosis GLM

Project Repository

- Used Python and R GLM packages to analyze NHANES 2023 data on Liver Steatosis using CAP score on 18 year olds with diet data
- Performed cumulative logistic regression with partial proportional odds to model cumulative probability of Liver Fibrosis categorization
- Performed HUM with 10-fold cross validation to analyze model performance
- Used residual analysis to identify outliers and refine model
- Tools Used: Python, R, Git

Indinavir Survival Analysis

Project Repository

- Analyzed the effect of Indinavir on the progression to AIDS in HIV positive patients with the ACTG 320 dataset
- Examined the assumptions of survival analysis and proportional hazards models and KM/NA survival curves
- Visualized and performed diagnostics on cox proportional hazards models
- Analyzed Time-Variied Coefficient Models and Parametric Models
- Tools Used: R, RStudio, Git, Beamer

Statistics Education Project

Confidential Data

- Analyzed and visualized demographics, self-efficacy, and confidence improvement data on an intro data-science courses students
- Used text analysis packages to create measures of free-response survey data to predict final exam grade in both data science and mathematical statistics courses
- Ran and optimized linear models to find significant predictors of final exam grades
- Collaborated with two other undergraduates to find solutions in my own and their projects
- Tools Used: R, RStudio, RShiny

ISI-BUDS Project

Project Repository

- Used mHealth data to create linear models and analyze trends in steps, heart rate, and presence of cardio-metabolic diseases in ethnic minority women
- Used multiple imputation with chained equations to account for missingness in time series data
- Used bootstrapping to create confidence intervals for linear regression coefficients
- Collaborated with two other undergraduates, a PhD. Students, and two professors to create a paper and presentation about mHealth and Cardiometabolic Disorders
- Tools Used: R, RStudio, MICE, kNN

Wikipedia Social Network Analysis

Wikipedia-SNA

- Collaborated with two other undergraduates to conduct social network analysis on Wikipedia article data provided by SNAP
- Created social network visualizations and metrics for hyperlinks (edges) between Wikipedia articles (nodes) to find correlations between edges and article popularity and detail
- Tools Used: R, RStudio, iGraph

Random Number Generation

RNG Project

- Used the Transformation, Accept-Reject, and Inverse CDF random number generation methods to create algorithms to randomly sample from Inverse Gamme, Log-Normal, and Weibull distributions respectively
- Used numerical and graphical inference to verify the validity of the sampling
- Tools Used: R, RStudio

Daily Nexus Article

Sinzenich to Santa Barbara

- Worked with copy editors to revise and learn the nuances of publishable writing prose
- Used MatPlotLib and Pandas to create data visualizations appropraite for all Daily Nexus readers
- Tools Used: Python, MatPlotLib, Jupyter

Additional Experience And Awards

Rama Thogarati Award Spring 2025; a monetary award given to six undergraduate students in the UCSB PSTAT Department for having the highest GPA in their graduating class

Floyd and Mary Schwall Fellowship Spring 2025; an internal UC Davis 21-month fellowship for incoming Ph.D. students focused on human medical research

Lurie Graduate Students Award Spring 2025; an internal UC Davis fellowship for Ph.D. students focused on human medical research

Referee of the Quarter Winter 2024; awarded soccer referee of the quarter for UCSB intramural soccer, helping in my promotion to head of officials

Technologies

Languages: Python, R, SQL, Base SAS, Latex, Bash

Software: RStudio, Rshiny, RSQLite, Jupyter, MySQL, SASOnDemand, GitHub

Packages: iGraph, MICE, kNN, matplotlib, Pandas, ggplot, Survival

Soft Skills: Time Management, Cooperation, Communication, Detail-Oriented, Funny if nothing else